QUANTUM COMPUTING

UNIT - II: Probability, Statistics, and Classical Mechanics Foundations

Detailed Lecture Notes with Color Figures and Key Equations

Prepared for classroom teaching and quick revision

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UNIT - II Overview

This unit builds the classical mathematical and physical intuition needed before formally introducing quantum postulates. Quantum states and measurement outcomes are probabilistic, so we first review probability and statistics. Quantum dynamics is most naturally written using Hamiltonians, so we also review Lagrangian and Hamiltonian mechanics and then connect classical Hamiltonians to quantum operators.

Learning Outcomes

- Compute and interpret descriptive statistics (mean, variance, standard deviation, covariance) for datasets and measurement outcomes.
- Apply probability rules (conditional probability, Bayes theorem, independence) to model uncertainty and inference.
- Work with random variables and common probability distributions used in modeling noise and measurement.
- Derive equations of motion using Lagrangian and Hamiltonian formulations for simple systems.
- Translate a classical Hamiltonian $H(q, p)$ into a quantum Hamiltonian operator $\hat{H}$ and relate it to unitary time evolution.

Notation used in these notes

Random variable: X (discrete or continuous)

Probability mass function (PMF): $P(X=k)$

Probability density function (PDF): $f_X(x)$

Expectation: $E[X]$ or □X□

Variance: $Var(X) = \sigma^2$

Generalized coordinates: q_i, generalized momenta: p_i

Planck constant: $\hbar = h/(2\pi)$

Quantum Hamiltonian operator: $\hat{H}$ (hat denotes operator)

1. Descriptive Statistics and Probability Basics

1.1 Types of data and why statistics matters in quantum measurement

In quantum experiments, the same preparation and measurement repeated many times produces a distribution of outcomes. Statistics provides the language to summarize these outcomes (location, spread, correlations) and to estimate unknown parameters such as a qubit state probability or a noise rate.

Data types

- Categorical (nominal/ordinal): labels such as detector click/no-click.

- Numerical: discrete counts (photon number) or continuous readings (voltage, time-of-flight).
- Population vs sample: in practice we observe a finite sample of measurement outcomes.

1.2 Measures of central tendency and dispersion

Given measurements x1, x2, ..., xN, common summaries are:

$$\text{Sample mean:}\quad \bar{x} = (1/N)\, \Sigma_{i=1}^{N} x_i \quad (1)$$

$$\text{Sample variance:}\quad s^2 = (1/(N-1))\, \Sigma_{i=1}^{N} (x_i - \bar{x})^2 \quad (2)$$

$$\text{Standard deviation:}\quad s = \sqrt{(s^2)} \quad (3)$$

$$\text{Covariance:}\quad Cov(X,Y) = E[(X - \mu_X)(Y - \mu_Y)] \quad (4)$$

$$\text{Correlation:}\quad \rho_{XY} = Cov(X,Y)/(\sigma_X \sigma_Y) \quad (5)$$

Interpretation tips

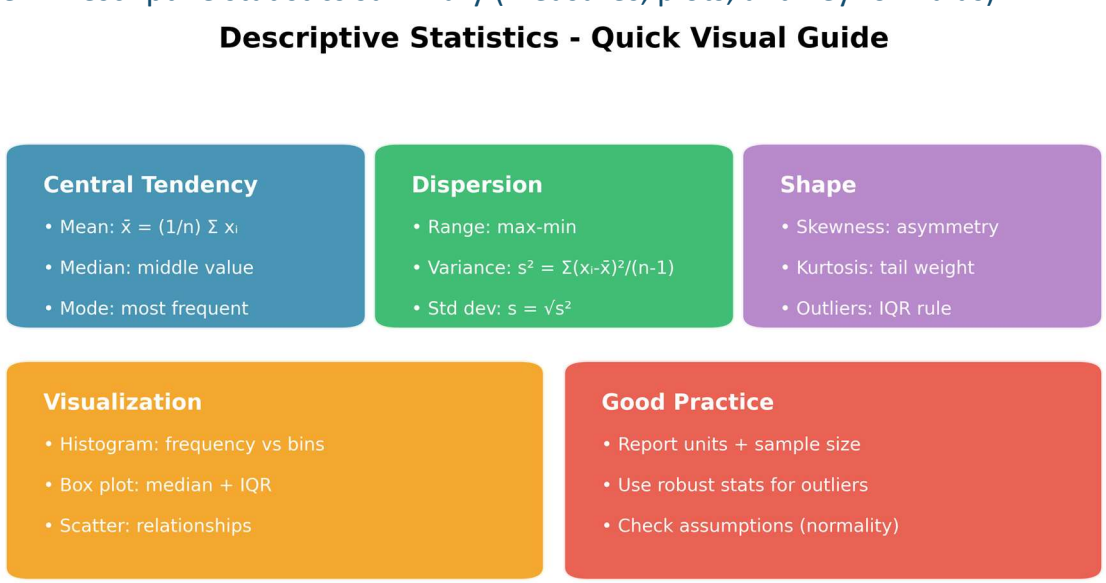
Mean ($\bar{x}$): "typical" value; sensitive to outliers.

Median: robust central value (50th percentile).

Variance/SD: spread/uncertainty around the mean.

Covariance/correlation: whether two variables move together (important in multi-qubit noise and readout).

Figure 1. Descriptive statistics summary (measures, plots, and key formulas).



Tip: For quantum experiments, always include uncertainty and error propagation when reporting measurements.

1.3 Probability: sample space, events, and axioms

Probability models uncertainty. A sample space Ω is the set of all outcomes. An event A is a subset of Ω. A probability measure P satisfies the axioms: (i) $0 \le P(A) \le 1$, (ii) $P(\Omega)=1$, and (iii) for disjoint events A_i, $P(\cup A_i)=\Sigma P(A_i)$.

Useful probability identities

$$P(A^c) = 1 - P(A) \quad (6)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (7)$$

$$\text{Conditional probability: } P(A|B) = P(A \cap B)/P(B), \; P(B)>0 \quad (8)$$

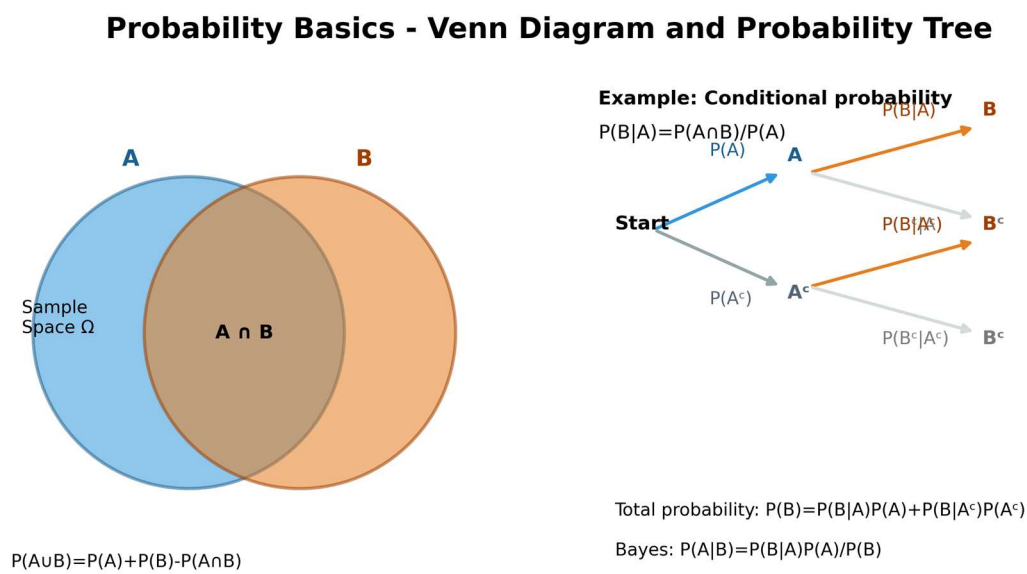
$$\text{Bayes theorem: } P(A|B) = P(B|A)\,P(A) / P(B) \quad (9)$$

$$\text{Law of total probability: } P(B) = \Sigma_i P(B|A_i)\,P(A_i) \quad (10)$$

Quantum connection (Born rule preview)

If a qubit is in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then measurement probabilities are $P(0) = |\alpha|^2$ and $P(1) = |\beta|^2$, with $|\alpha|^2 + |\beta|^2 = 1$.

Figure 2. Venn diagram and a probability tree for conditional probability and Bayes reasoning.



2. Random Variables and Common Distributions

2.1 Random variables, PMF/PDF, and CDF

A random variable (RV) X assigns a numerical value to each outcome in Ω. Discrete RVs (counts, bit outcomes) use a probability mass function (PMF) P(X=k). Continuous RVs (time, voltage) use a probability density function (PDF) f_X(x).

$$\text{CDF: } F_X(x) = P(X \leq x) \quad (11)$$

$$\text{For continuous X: } P(a \leq X \leq b) = \int_\{a\}^\{b\} f_X(x)\, dx \quad (12)$$

2.2 Expectation, variance, and moments

Expectation summarizes the average value over many trials:

$$\text{Discrete: } E[X] = \Sigma_k\, k\, P(X=k) \quad (13)$$

$$\text{Continuous: } E[X] = \int_\{-\infty\}^\{\infty\} x\, f_X(x)\, dx \quad (14)$$

$$\text{Variance: } Var(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2 \quad (15)$$

$$\text{Standard deviation: } \sigma = \sqrt{Var(X)} \quad (16)$$

$$\text{Moment generating function (MGF): } M_X(t) = E[e^\{tX\}] \quad (17)$$

Rules that save time

Linearity: E[aX + b] = aE[X] + b
Var(aX + b) = a^2 Var(X)
If X and Y are independent: E[XY] = E[X]E[Y]
Law of total expectation: E[X] = E[E[X|Y]]

2.3 Common distributions used in measurement and noise modeling

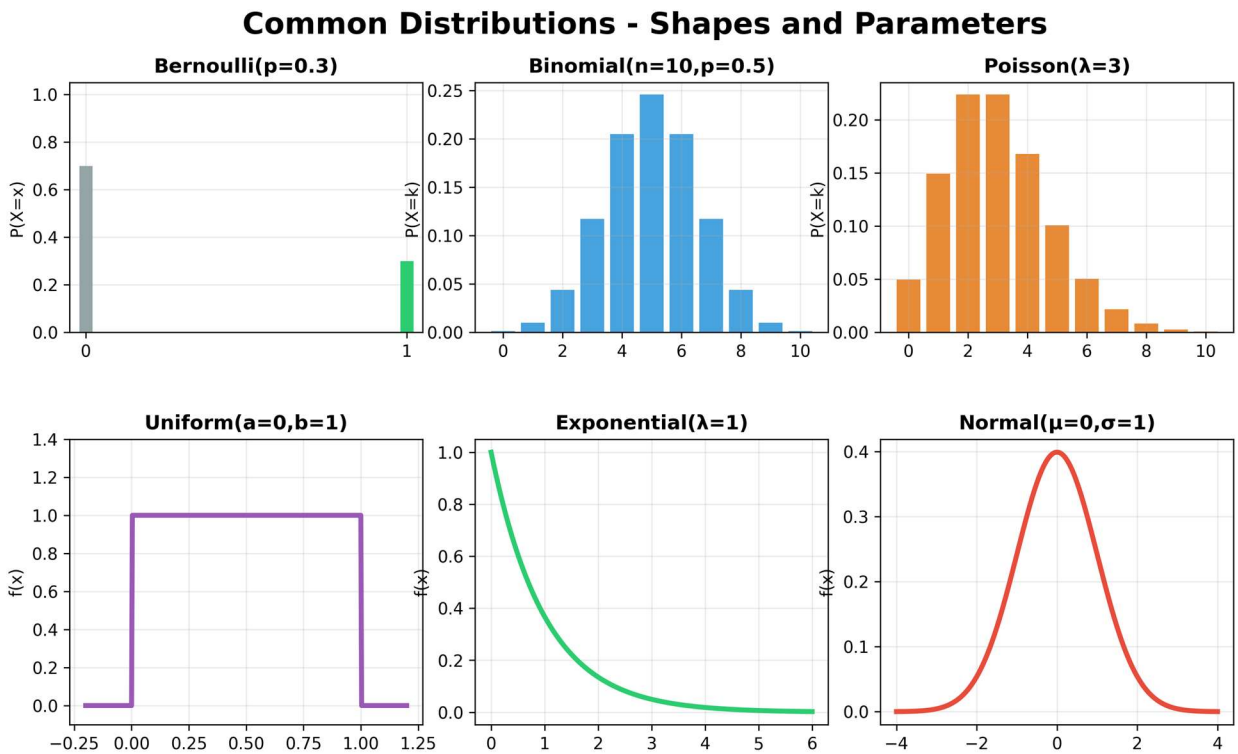
In quantum computing labs, distributions appear when counting events (Poisson), modeling bit-flip errors (Bernoulli/Binomial), or describing aggregated noise (Normal via central limit theorem).

Summary table: PMF/PDF, mean, and variance

Distribution	Parameters	PMF/PDF	Mean (μ)	Variance (σ^2)
Bernoulli	p	P(X=1)=p, P(X=0)=1-p	p	p(1-p)
Binomial	n, p	P(X=k)=C(n,k)p^k(1-p)^{n-k}	np	np(1-p)
Poisson	λ	P(X=k)=e^{-λ} λ^k / k!	λ	λ
Uniform	a, b	f(x)=1/(b-a), a≤x≤b	(a+b)/2	$(b-a)^2/12$

Exponential	λ	f(x)=λ e^{-λx}, x≥0	1/λ	$1/\lambda^2$
Normal	μ, σ	f(x)=(1/(σ√(2π))) e^{-(x-μ)²/(2σ²)}	μ	σ^2

Figure 3. Shapes and parameters of common probability distributions.



2.4 Central Limit Theorem (CLT) and why Gaussians appear everywhere

The CLT states that sums (or averages) of many independent, identically distributed random variables tend to a normal distribution even when the original variables are not normal (under mild conditions). In experiments, many small noise contributions add up, so readout errors, timing jitter, and analog noise often look approximately Gaussian.

$$\text{If } S_N = (X_1 + \ldots + X_N), \text{ then } (S_N - N\mu)/(\sigma\sqrt{N}) \to \text{Normal}(0,1) \text{ as } N\to\infty \quad (18)$$

3. Classical Mechanics Foundations

3.1 Lagrangian formulation

Classical mechanics can be written using generalized coordinates q_i that automatically incorporate constraints. Define the Lagrangian as $L(q, \dot{q}, t) = T(q, \dot{q}, t) - V(q, t)$, where T is kinetic energy and V is potential energy.

$$\text{Action: } S[q(t)] = \int_{t1}^{t2} L(q, \dot{q}, t)\, dt \quad (19)$$

$$\text{Principle of least action: } \delta S = 0 \quad (20)$$

$$\text{Euler-Lagrange equation:}\quad d/dt(\partial L/\partial \dot{q}_i) - \partial L/\partial q_i = 0 \quad (21)$$

How to use Euler-Lagrange (recipe)

1) Choose generalized coordinate(s) q_i.
2) Write T and V, then L = T - V.
3) Compute $\partial L/\partial q_i$ and $\partial L/\partial \dot{q}_i$.
4) Apply $d/dt(\partial L/\partial \dot{q}_i) - \partial L/\partial q_i = 0$ to obtain equations of motion.

Worked Example 1: Simple harmonic oscillator (mass-spring)

Let q = x be displacement, $T = (1/2)\, m\, \dot{x}^2$, $V = (1/2)\, k\, x^2$.

$$L = (1/2)\, m\, \dot{x}^2 - (1/2)\, k\, x^2 \quad (22)$$

Compute derivatives: $\partial L/\partial x = -kx$, $\partial L/\partial \dot{x} = m\, \dot{x}$.

$$d/dt(m\, \dot{x}) + kx = 0 \;\Rightarrow\; m\, \ddot{x} + kx = 0 \quad (23)$$

The solution is sinusoidal motion with angular frequency $\omega = \sqrt{(k/m)}$.

$$x(t) = A\cos(\omega t) + B\sin(\omega t) \quad (24)$$

3.2 Hamiltonian formulation

The Hamiltonian formulation is particularly important because quantum time evolution is generated by a Hamiltonian operator. We first define generalized momentum p_i conjugate to q_i:

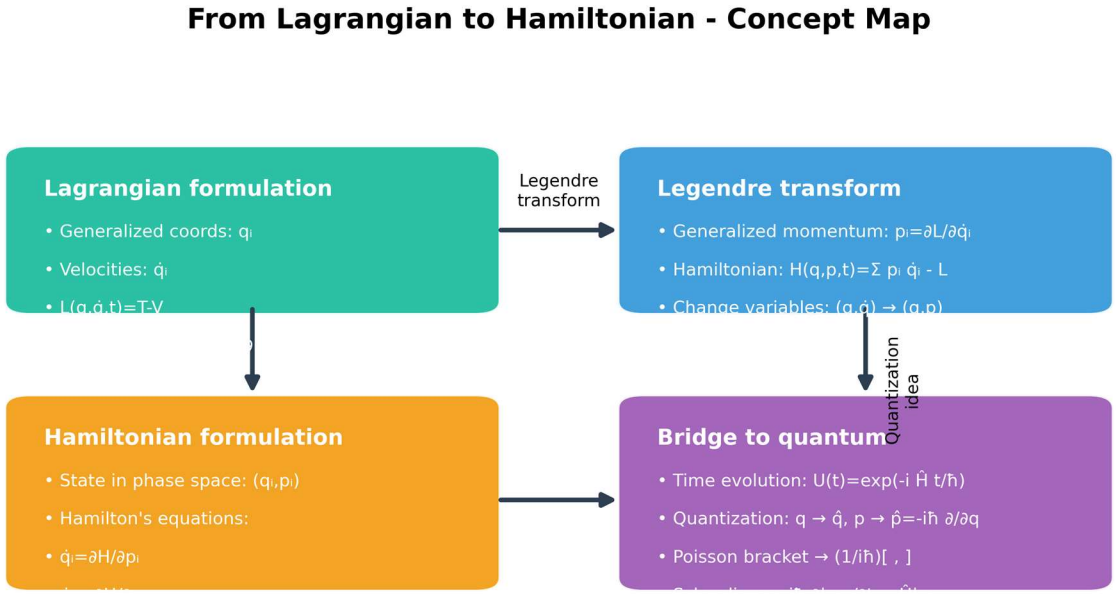
$$p_i = \partial L/\partial \dot{q}_i \quad (25)$$

Then define the Hamiltonian as a Legendre transform of L:

$$H(q,p,t) = \Sigma_i\, p_i\, \dot{q}_i - L(q, \dot{q}, t) \;\; (\text{with } \dot{q} \text{ expressed in terms of } q,p) \quad (26)$$

$$\text{Hamilton's equations:}\quad \dot{q}_i = \partial H/\partial p_i, \quad \dot{p}_i = -\partial H/\partial q_i \quad (27)$$

Figure 4. From Lagrangian to Hamiltonian: Legendre transform and flow of ideas.



Worked Example 2: Hamiltonian of the harmonic oscillator

From $L = (1/2)\, m\, \dot{x}^2 - (1/2)\, k\, x^2$, compute momentum $p = \partial L/\partial \dot{x} = m\, \dot{x}$.

$$\dot{x} = p/m \quad (28)$$

$$H = p\,\dot{x} - L = p(p/m) - [(1/2)m(p/m)^2 - (1/2)k\,x^2] \quad (29)$$

$$H(x,p) = p^2/(2m) + (1/2)\,k\,x^2 \quad (30)$$

Hamilton's equations give $\dot{x} = p/m$ and $\dot{p} = -kx$, which reproduces $m\,\ddot{x} + kx = 0$.

3.3 Phase space and Poisson brackets

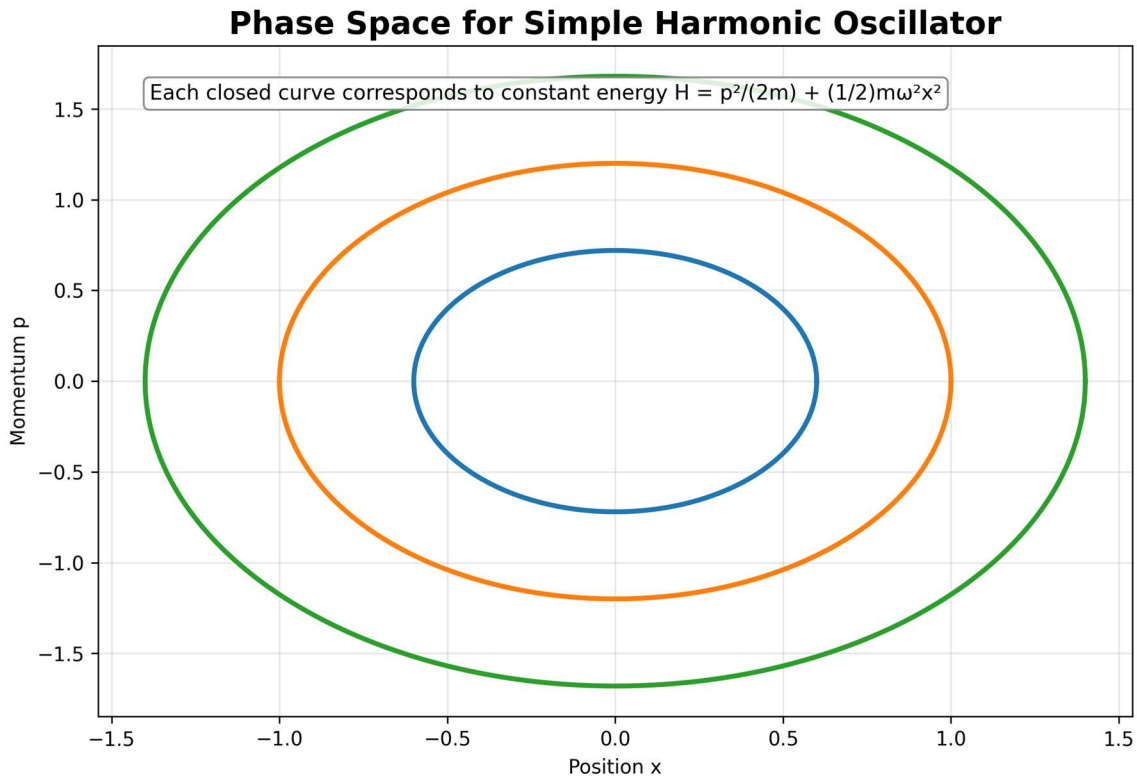
In Hamiltonian mechanics, the system state is a point in phase space (q,p). The time evolution traces a curve in phase space. For two functions f(q,p) and g(q,p), the Poisson bracket is:

$$\{f,g\} = \Sigma_i\, (\partial f/\partial q_i)(\partial g/\partial p_i) - (\partial f/\partial p_i)(\partial g/\partial q_i) \quad (31)$$

Time evolution of any observable f(q,p,t):

$$df/dt = \{f,H\} + \partial f/\partial t \quad (32)$$

Figure 5. Phase space trajectory for a simple harmonic oscillator (constant energy ellipse).



4. Relation Between Classical Hamiltonians and Quantum Operators

4.1 Correspondence principle and canonical quantization

The correspondence principle states that quantum mechanics should reproduce classical mechanics in the appropriate limit (large quantum numbers or ħ→0). A common route from classical to quantum is canonical quantization: promote classical variables to operators and replace Poisson brackets with commutators.

$$q_i \to \hat{q}_i, \quad p_i \to \hat{p}_i \quad (33)$$

$$\text{Canonical commutation:} \quad [\hat{q}_i, \hat{p}_j] = i\hbar\, \delta_{ij} \quad (34)$$

$$\text{Poisson bracket correspondence:} \quad \{f,g\} \to (1/(i\hbar))[\hat{f}, \hat{g}] \quad (35)$$

Position representation (most common in 1D):

$$\hat{x}\, \psi(x) = x\, \psi(x) \quad (36)$$

$$\hat{p}\, \psi(x) = -i\hbar\, (d/dx)\, \psi(x) \quad (37)$$

4.2 From Hamiltonian function to Hamiltonian operator

Given a classical Hamiltonian $H(x,p) = p^2/(2m) + V(x)$, the corresponding quantum Hamiltonian operator is obtained by substituting p with the momentum operator. This yields:

$$\hat{H} = \hat{p}^2/(2m) + V(\hat{x}) \quad (38)$$

$$\text{In x-representation: } \hat{H} = -(\hbar^2/(2m))\, d^2/dx^2 + V(x) \quad (39)$$

Quantum dynamics is generated by $\hat{H}$ through the Schrödinger equation:

$$i\hbar\, \partial|\psi(t)\rangle/\partial t = \hat{H}\, |\psi(t)\rangle \quad (40)$$

$$\text{Unitary evolution: } |\psi(t)\rangle = U(t)\, |\psi(0)\rangle, \quad U(t)=\exp(-i\, \hat{H}\, t/\hbar) \quad (41)$$

Why this matters for quantum gates

Single- and two-qubit gates are unitary operators.

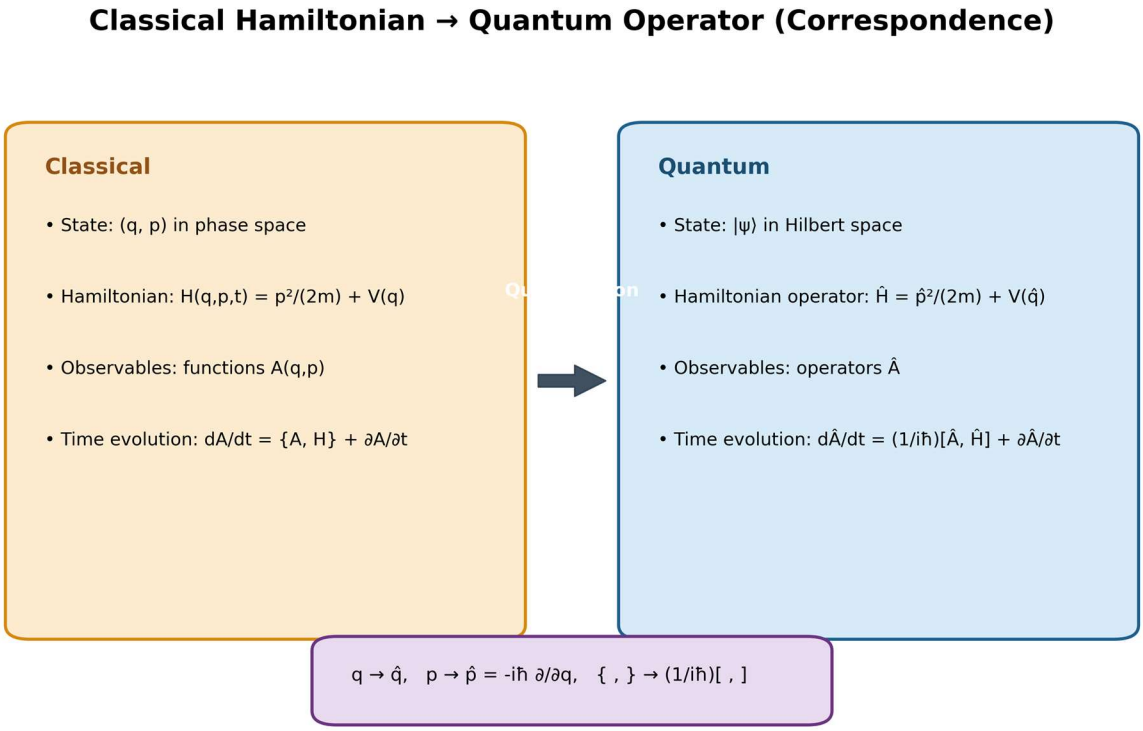
In many physical platforms, gates are implemented by applying a controlled Hamiltonian for a fixed time.

Example: if $\hat{H} = (\hbar\Omega/2)\, \sigma_x$, then $U(t) = \exp(-i\, \Omega\, t\, \sigma_x/2)$ is an X-rotation on the Bloch sphere.

4.3 Operator ordering and practical cautions

Not every classical expression has a unique quantum operator because operators may not commute. For example, $xp \neq px$ as operators. When quantizing terms like x p x or p x p, one must choose an ordering (often symmetric/Weyl ordering) guided by physics, symmetry, and experiments.

Figure 6. Summary map: classical variables, Poisson brackets, and the corresponding quantum operators/commutators.



5. Short Applications to Quantum Computing

5.1 Noise as random variables (bit-flip example)

Suppose a qubit experiences a bit-flip error with probability p during a short time step. Define X as a Bernoulli RV where X=1 indicates an error. Then E[X]=p and Var(X)=p(1-p). After n independent time steps, the number of flips K follows a Binomial(n,p) distribution.

$$P(K=k) = C(n,k)\, p^k\, (1-p)^{n-k} \quad (42)$$

5.2 Measurement outcomes and estimation

If we measure a qubit in the computational basis repeatedly and observe m outcomes of |1□ out of N trials, the estimator of P(1) is $\hat{p} = m/N$. The standard error (approx.) for a Bernoulli process is $\sqrt{\hat{p}(1-\hat{p})/N}$. Larger N reduces estimation uncertainty.

$$\hat{p} = m/N \quad (43)$$

$$SE(\hat{p}) \approx \sqrt{\hat{p}(1-\hat{p})/N} \quad (44)$$

6. Quick Revision Sheet

Must-remember formulas (one page)

Conditional probability: $P(A|B) = P(A\cap B)/P(B)$
Bayes: $P(A|B) = P(B|A)P(A)/P(B)$
Expectation: $E[X] = \Sigma\, k\, P(X=k)$ or $\int x\, f(x)\, dx$
Variance: $Var(X) = E[X^2] - (E[X])^2$
Euler-Lagrange: $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$
Hamilton: $\dot{q} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial q$
Quantization: $p \to -i\hbar\, d/dx$, $[x,p]=i\hbar$
Schrödinger: $i\hbar\, \partial|\psi\rangle/\partial t = \hat{H}|\psi\rangle$, $U=\exp(-i\hat{H}t/\hbar)$

7. Practice Questions (with short answers)

1. Q1. A dataset has values 2, 4, 4, 4, 5, 5, 7, 9. Compute mean and sample variance.

   Answer: Mean = 5.0. Sample variance $s^2 = 4.571$ (approx.).

2. Q2. If $P(A)=0.4$, $P(B)=0.5$ and $P(A\cap B)=0.2$, compute $P(A|B)$ and $P(B|A)$.

   Answer: $P(A|B)=0.2/0.5=0.4$; $P(B|A)=0.2/0.4=0.5$.

3. Q3. A qubit readout has error probability $p=0.02$ per shot. In $N=200$ shots, what is the expected number of wrong outcomes?

   Answer: $E[K]=Np=200*0.02=4$.

4. Q4. Write the Lagrangian for a free particle of mass m and derive the equation of motion.

   Answer: $L=(1/2)m\, \dot{x}^2$; Euler-Lagrange gives $\ddot{x}=0$.

5. Q5. For $H(x,p)=p^2/(2m)+V(x)$, write the quantum Hamiltonian operator in x-representation.

   Answer: $\hat{H}=-(\hbar^2/(2m))\, d^2/dx^2 + V(x)$.

6. Q6. State the correspondence between Poisson brackets and commutators.

   Answer: $\{f,g\} \to (1/(i\hbar))[\hat{f},\hat{g}]$.

7. Q7. A Binomial(n=10,p=0.3) variable K: compute P(K=0).

   Answer: P(K=0)=(1-p)^n=0.7^{10}.

8. Q8. An exponential RV with rate λ has mean $1/\lambda$. If mean is 2 seconds, find λ.

   Answer: $\lambda=0.5$ s^{-1}.

9. Q9. Given $L=(1/2)m\,\dot{x}^2-(1/2)k\,x^2$, compute p and H.

   Answer: $p=m\dot{x}$; $H=p^2/(2m)+(1/2)kx^2$.

10. Q10. If $\hat{H}=(\hbar\Omega/2)\sigma_x$, write U(t).

    Answer: $U(t)=\exp(-i\Omega t\,\sigma_x/2)$.